Test Plan – Swag Labs

This is the test plan for Swag Labs (https://www.saucedemo.com), a demo e-commerce web application.

Component	Details
Objectives	The goal is to ensure the core functionality of Swag Labs (Login, Products, Cart, Checkout) performs as expected and meets UI, performance, usability, and security standards. Specific objectives include: • Execute 100% of Critical/High test cases, with all Critical test cases passing and every failed test documented in a defect report • Verify sorting and pricing (Item total + 8% Tax = Total) with zero calculation errors • Verify that the core flows (Login, Products, Cart, Checkout) work on the latest versions of Chrome, Firefox, and Edge, and on a mobile viewport using Chrome DevTools emulation • Automate a small smoke/regression suite covering the main flow (Login, Add to Cart, Checkout) to speed up retesting
Approach	The test strategy combines **manual and automated testing**. • **Manual testing** will be used for functional, usability, exploratory, and UI/visual testing (especially visual_user). • **Automated testing** will cover a small smoke/regression suite for the main flow (Login, Add to Cart, Checkout) using Selenium WebDriver with TestNG. Test cases suitable for automation are marked in the test case spreadsheet. • Test cases are documented in a spreadsheet; manual results are tracked manually and automated results are generated by the test framework. • Functional, compatibility, performance (page load via Lighthouse), and basic security testing (direct URL access without login, session after logout, checkout input validation).
Scope	**In Scope:** • User authentication (valid, invalid, locked_out_user, empty fields, Logout) • Products page and sorting (Name A–Z/Z–A, Price low–high/high–low) • Product details page • Shopping cart (Add, Remove, badge, persistence) • Checkout (Information, Overview, Complete) and field validation • Special users: problem_user, performance_glitch_user, error_user, visual_user • Burger menu (All Items, About, Logout, Reset App State) • Test automation of a small smoke/regression suite for the main flow **Out of Scope:** • Automation of exploratory and visual testing (visual_user stays manual) • Real payment gateway (demo site) • Backend/database testing, heavy load/stress testing • Native mobile apps and advanced penetration testing
Test Deliverables	• Test cases (manual and automated) • Automation scripts • Automated test execution reports • Defect reports • Test execution report • Final test summary report These deliverables will be shared weekly with stakeholders and archived for future regression testing.
Dependencies	• Availability of https://www.saucedemo.com (external service) • Stable internet connection • Latest browser versions • Automation setup (Java, Maven, Selenium WebDriver, TestNG) **Contingency plan:** if the site is down, continue test case design and review, and postpone execution.

Component	Details
Test Environment	• Browsers: Chrome, Firefox, Edge (latest version) • Mobile: Chrome DevTools emulation • Network: Wi-Fi and throttled 3G/4G • Tools: Browser DevTools, Lighthouse, Jira, Selenium WebDriver, TestNG • Automation runs: Chrome
Risk Management	**Identified Risks:** • Site unavailable or changed without notice • Intentional glitches in problem/error/performance users cause confusion • Missed test cases or human error in manual execution • Flaky automated tests caused by timing issues or the intentional delays of performance_glitch_user • Limited automation experience in the team **Mitigation Strategies:** • Daily smoke test of the core flows • Document expected behavior of special users vs real bugs • Peer review of test cases and execution results • Use explicit waits, and automate only the stable main flow
Schedule	**Timeline:** • Test Planning (Test Plan delivery): Fri Oct 30, 2026 • Test Case Design: Date TBD (2 days) • Functional Testing: Date TBD (2 days) • Automation Script Development (parallel with functional testing): Date TBD (2 days) • Compatibility, Performance and Security Testing: Date TBD (2 days) • Automated and Manual Regression and Retest: Date TBD (1 day) • Test Reporting and Sign-off: Date TBD (1 day)
Roles and Responsibilities	• Test Engineer: Ziad Mohammed Mahmoud Abdo • Test Engineer: Haia Esam Ali Farghaly • Test Engineer: Mahmoud Ahmed Abdulhakeem Ahmed • Test Engineer: Tarek Sayed Mahmoud Mohammed • Test Engineer: Peter Ayman Mahrous Fahmy • Test Engineer: Abdulrahman Eid Yassin Bayoumi